

A Lean Approach to Estimate the Functional Size of Operating Applications

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Abstract—Functional size measures have been gaining increasing acceptance not only for project estimation and productivity comparisons but also for the evaluation of the company assets concerning with operating applications. However, functional sizing operating applications can represent a complex task (they could be developed in past projects, in some cases by others, with long standing technologies, and without useful documentation supporting them). Thus, companies require cheaper strategies to estimate functional size of these applications. To this end, we proposed some measurement approaches concerning with the number of physical tables and the number of read and read/written tables that can be easily collected from the database schema and the application interfaces. We assessed the effectiveness of the proposals performing an empirical study based on 17 applications of an Italian company. The results highlight that the proposed approaches allowed to obtain accurate functional size estimations, thus turning out to be suitable for the involved company.

Keywords- *Software functional size measurement, Function Points, Physical tables, Operating applications*

I. INTRODUCTION

Software sizing is crucial during the whole software development process since it supports other project management activities (e.g., estimating, allocating, or tracking) [15]. Among the measurement approaches proposed in the literature, Functional Size Measurement (FSM) is widely employed to measure the size of Users Functional Requirements [20]. The first FSM method was Function Point Analysis (FPA) proposed by Albrecht [2]. Function Points are widely adopted since they are independent from the non-functional technical and quality requirements [48]. Furthermore, it is an ISO standardized technique [24] and some organizations (e.g. IFPUG [22] and NESMA [41]) works around the world to improve and promote the application of the approach.

Measuring the size of operating software applications is also crucial for companies when they want to have a clear “quantification” of the company software applications stock for several purposes, e.g., to support the decisions of the board of directors, to provide attestations with stakeholders, etc... This measurement is considered a complex task since

the company stock can contain applications developed in several past projects that evolved several times, possibly with several development processes, in many cases developed by others, with long standing technologies, and without useful documentation supporting them. Thus, it is often hard to find people of the company that have a complete view of the whole application. As a consequence, to carry out the measurement, it is necessary to contact/interview several people to get information and partial and inconsistent answers could be obtained. As a result an undefined level of uncertainty is introduced in the measurement that is based just on the information specified by the involved people. Furthermore, the measurement of operating applications can also be very expensive since the companies have to involve experts of the specific functional size measurement method.

Thus, it could be very useful to identify an approach able to estimate the functional size of operating applications using specific technical components, easily available to a project manager without employing further resources. It could be argued that the most simple way to do this might be to count the number of Lines Of Code (LOC) and then use a backfiring approach [26] to derive an estimation of the functional size of the application. Nevertheless, how highlighted by the practitioners of the software company involved in our study and how it is also reported in the literature [14] LOC and backfiring present several drawbacks due to the use of different platforms and programming languages and the lack of a standard way to do the counting.

Thus, in this paper we present simple sizing approaches to be applied on operating applications based on specific software components, such as the number of physical tables and the number of read and read/written tables that can be obtained by the technical responsible of the software application by exploiting the database schema and the application interfaces. This information is that employed by measurers to determine the Base Functional Components (BFCs) when a functional size measurement method, like IFPUG Function Points or COSMIC, is applied to determine the size of an application. The idea motivating our research is similar to the one behind investigations proposing the use

of single BFCs to provide estimates of software development effort as soon as possible during the first phases of the software life cycle (e.g., [8][9][19]).

Then, we carried out an empirical study to assess the proposed approaches for estimating the functional size of operating applications given in terms of IFPUG Function Points [22].

In particular, the research questions addressed in this paper are:

RQ1: *Can the number of physical tables be exploited to predict the application size (in terms of Function Points)?*

RQ2: *Can the number of read and read/written tables be exploited to predict the application size (in terms of Function Points)?*

To perform the empirical analysis we have exploited data from 17 applications managed by an Italian company. As for the technique, we employed linear regression analysis using a leave-one-out cross validation. In order to assess the accuracy of the obtained size estimations we have employed summary measures like MMRE, MdMRE, and Pred(25), which are widely used evaluation criteria. Boxplot of residuals and statistical test have been also used to compare the accuracy of the estimations achieved with the different prediction models.

The paper is organized as follows: Section 2 discusses related work while Section 3 introduces the research method used in this paper. Section 4 presents and discusses the results from the statistical analysis carried out on the employed industrial sample. Section 5 discusses threats to validity while Section 6 provides conclusions of the current work and suggests future research directions.

II. RELATED WORK

To the best of our knowledge no previous work has exploited information on physical tables and the number of read and written tables to estimate functional size of applications. The works proposing the backfiring approach can be considered related to our investigation since the motivations underlying the two researches are quite close, e.g., lack of time and budget to count Function Points and few (or no) documentation available on software requirements to apply Function Points Analysis [14]. Since the initial proposal of Kaper Jones [26], the backfiring approach has been widely applied in practice (e.g., [21][27][37]). However, some papers have highlighted issues in the application of backfiring, mainly related to the LOC and due to the difficulties rising from the use of different platforms and programming languages for the applications to be measured and the lack of a standard way to do the counting [14][43][50].

Other researches that can be related to our work are those proposing and/or investigating the use of approximate or simplified sizing approaches and those about the use of single Base Functional Components (BFCs) to provide a size estimation of the application under development as soon as possible during the first phases of the software life cycle.

Two widely known approaches for simplifying the application of Function Points Analysis are the “Early & Quick Function Points (EQFP)” [46] and simplified version of NESMA [41]. The idea behind the application of EQFP is based on the fact that some estimates are required before the requirements analysis is completed. Thus, some information is not available to fully apply FPA and EQFP relaxes the application of the FP method by not considering some details. The method can also be applied when a complete requirements documentation is available but the managers intend to complete the measurement in less time and with a lower cost (i.e., admitting a trade-off between a correct measurement and time and budget available). In particular, data and processes of an application are classified in several groups according to a hierarchy [46], where some groups contain the BFCs of FPA and other groups are aggregation of Base Functional Processes (e.g., Macro Process, General Process) or Data Groups (e.g., Multiple Data Group). The idea of the approach is that different parts of the software can be measured at the different level of details. Thus, exploiting functional components more coarse-grained than the BFCs of FPA, EQFP approach produces an approximation of the size in terms of Function Points.

The first approximation suggested by NESMA (named Indicative NESMA) is based on the identification of Internal Logical Files (ILF) and External Interface Files (EIF), i.e., without exploiting the contribution of Transactions Functions (i.e., EI, EQ, and EO) of FPA, and applying predefined weights [3]. The second approximation proposed by NESMA (named Estimated NESMA) exploits all the BFCs of FPA, but the complexity of the identified components are not determined. By default, a low complexity is associated to ILF and EIF elements while a medium complexity is considered for EI, EQ, and EO.

The above approaches have been applied in [3], where a model based measurement is combined with the application of the NESMA simplified measurement processes. In particular, the software to be measured is first specified in terms of UML documents providing guidelines that allow designer to model all the information required to apply FPA. Then, the proposed measurement method allows to map the UML elements on the BFCs of FPA and obtain the approximate size.

EQFP and the two simplified NESMA have been also applied to size both “traditional” and real-time applications for evaluating their accuracy with respect to FPA [34]. The results of the performed analysis show that EQFP was more accurate in size estimation of non real-time applications, while NESMA approaches turned out to be quite good for both applications types.

An initial proposal, very close to EQFP has been presented in [36], where the Early Function Points (EFP) and Extended Function Points (XFP) methods have been presented with the purpose of obtaining the amount of Function Points in the early phases of a software development life cycle reducing time and cost. EFP just suggests to apply FPA with the available information obtaining estimations that are compatible with IFPUG 4.0 Function Points. XFP extends EFP by applying correction

factors, based on the level of reuse and on a classification of algorithms in terms of their complexity. EXP is not compatible with IFPUG 4.0 Function Points but in the study carried out in [36] was able to produce better size estimates than EFP.

In [42] two size metrics based on information obtained from use cases (i.e., transactions and entity objects) are presented and used for effort estimation in two empirical studies. The first study is based on information coming from four academic projects, while the second one employs data from four industrial projects. The goal of the performed analysis was to compare the estimations obtained with these two metrics. The results show the use of transactions as a size metric can provide better effort estimation than those obtained with entity objects.

Some studies have analyzed the different contribution of BFCs against the overall 'fsu' value (e.g., UFP for IFPUG [24] or CFP for COSMIC [11]) in estimating software development effort. In particular, two empirical studies based on ISBSG samples [23] showed that linear regression models employing two or more BFCs (e.g., EI, EO, EQ, ILF, EIF for the IFPUG method and Entry, eXit, and Read, Write for the COSMIC method) as independent variables instead of the whole single fsu value (e.g., UFP for the IFPUG method or CFP for the COSMIC method) were characterized by more accurate estimations [8], [19]. The study presented in [17] and its replication described in [9] are further applications of the idea proposed in [8], [19], by exploiting a data set of Web applications developed by a single company operating in Italy. The results of the performed empirical analysis confirmed that the use of combinations of two or more BFCs can provide better estimation of development effort.

III. RESEARCH METHOD

In the following, a description of the industrial data set used in this study as well as the analysis procedure performed to assess our research questions are presented.

A. Data set description

In our analysis we employed a data set provided by Infocamere¹, the ICT consortium company responsible for the design and management of the network of the Italian Chambers of Commerce nationwide. Its statutory mandate is the management of its key financial registers (register of companies, register of non payments, statements of accounts etc.). The customer base consists of 105 Italian Chambers of Commerce, professional associations, trade associations, local and central public administrations, business and economic information agencies. In particular, InfoCamere has developed and manages the national system of data transmission connecting the 105 Chambers of Commerce and their 300 branch offices via a high speed, high security network. It is also the national certifying Authority issuing digital certificates for Tachograph Cards. The company designs, develops and delivers services via a Management

System that complies with ISO 9001, having initially been certified in 1997 (ISO 9001:1994), then approved for Vision 2000 (ISO 9001:2000) and finally extended to the latest version (ISO 9001:2008) in March 2009.

The company provided us information on a set of 17 applications consisting of management information systems (50%) and applications mainly characterized by query and reporting systems (50%). These applications were developed between 2005 and 2010 and can be considered quite homogeneous in terms of adopted technologies (Java, Asp, PL/SQL), and development teams. No selection of the applications was done.

The information we exploited in our empirical analysis are: application size, in terms of number of Function Points, the number of physical tables, and the number of read tables and the number of read/written tables. The number of Function Points refers to the result of the application of IFPUG measurement method to the deployed software product. This measurement task is included in the software life cycle for each project developed by the company. This information contributes to the productivity database of the company, together with the information on the development effort, and it is exploited for internal benchmarking purposes (to verify the company trend year by year) and by project managers for providing initial estimates for new projects. The number of physical tables has been obtained from the database schema and the application interfaces of the considered operating applications. The available documentation allowed us to collect the number of read tables and the number of read/written tables. We did not use all the three measures together since the number of read tables and the number of read/written tables are obtained by further classifying the physical tables. Data collection was made by the same person of the company to improve consistency among measurements.

Table I reports the descriptive statistics of the variables FP, denoting the application size in terms of number of Function Points; T, denoting the number of physical tables; R, denoting the number of tables that are read; and R/W, denoting the number of tables that are read and written, for the applications considered in our study².

TABLE I. DESCRIPTIVE STATISTICS OF THE EMPLOYED DATA SET

Entity	Obs	Min	Max	Mean	Median	Std.Dev
FP	17	78	710	234.7	150	182.42
T	17	7	69	24.94	17	18.98
R	17	3	56	19.18	8	19.68
R/W	17	3	21	0.82	8	5.78

B. Analysis procedure

The goal of our study is to verify whether or not the number of physical tables (and the number of read and written

¹ www.infocamere.it

² For confidentiality reasons, it's not possible to provide details from the projects but only the high-level figures.

tables) can be exploited to predict the application size (in terms of Function Points). To this end, we first verified the strength of the relationship between FP and T and between FP and R and R/W. In particular, we exploited Linear Regression (LR) and Manual Stepwise Regression (SWR). LR is one of the most commonly used statistical techniques for exploring the relationship between a dependent variable and one or more independent variables, providing a prediction model described by an equation [35]:

$$y = b_1x_1 + b_2x_2 + \dots + b_nx_n + c \quad (1)$$

where y is the dependent variable (the size), $x_1, x_2, \dots, x_n$ are the independent variables (the predictors) with coefficient b_i , and c is the intercept. In our empirical study we have exploited LR to obtain linear regression models that use FP as dependent variable and T as independent variable.

For R and R/W, we applied SWR that allows us to obtain estimation models by adding, at each step, the independent variable with the highest association to the dependent variable, taking into account all the variables (i.e., R and R/W) currently in the model. To select the variables to be added in the model a Manual SWR (MSWR) can be applied, using the technique proposed by Kitchenham in [28]. The idea underlying this procedure is to find the set of independent variables that better explain the variation in the dependent variable, and then to use linear regression to obtain the final model.

To evaluate the goodness of fit of a regression model, several indicators can be considered. Among them, the square of the linear correlation coefficient, R^2 , shows the amount of the variance of the dependent variable explained by the model related to the independent variable. Other useful indicators are the F value and the corresponding p-value (denoted by Sign F), which high and low values, respectively, denote a high degree of confidence for the prediction. We have also considered the p-values and t-values for the corresponding coefficients and the intercept. The p-values give an insight into the accuracy of the coefficients and the intercept, whereas their t-values allow us to evaluate their importance for the built model. In particular, p-values less than 0.05 are considered an acceptable threshold, meaning that the variables are significant predictors with a confidence of 5%. As for the t-value, a variable is significant if its value is greater than 1.5.

It is worth noting that whenever variables are highly skewed they can be transformed before being used in the linear regression. This is done to satisfy the assumptions underlying linear regression analysis [35]. A widely used transformation is the one considering the natural log (Ln), which makes larger values smaller and brings the data values closer to each other [35]. In addition, whenever a variable needed to be transformed but had zero values, the natural logarithmic transformation was applied to the variables value after adding 1 as done in [38]. Moreover, the stability of each model built using linear regression should be verified. In order to accomplish this step, we can use a

residual plot showing residuals vs. fitted values to investigate if the residuals are randomly and normally distributed [35]. Then, Cook's distance values can be used to identify the influential data points. As suggested in [38] each point having distances higher than $3 \times (4/n)$, where n represents the total number of projects, are immediately removed from the data analysis. On the other hand, the points having distances higher than $4/n$ but smaller than $(3 \times (4/n))$ are removed and the stability of model is tested by analyzing the effect of their removal on the model. If the goodness of fit improves (i.e., a higher R^2 is obtained and the coefficients of the model remain stable then the influential data points are not excluded from the analysis.

Moreover, to validate the accuracy of the obtained size estimation models (i.e., to verify whether or not the predicted sizes were useful estimations of the actual sizes) we exploited a leave-one-out cross validation. It is widely used in empirical studies when dealing with small data sets [4]. In particular, to apply the technique, the original data set is divided into N different subsets (where N is the size of the original data set) of training and validation sets, where each validation set has one data point. Then, N steps are performed and at each step, the training set is used to determine the estimation model and the validation set to assess the obtained estimates. The accuracy of the obtained estimations is assessed by using summary measures MMRE, MdMRE and Pred(l) [10], which have been widely used in empirical studies to assess the accuracy of software development effort estimation models (see e.g., [5], [13], [16], [25], [29], [38], [39], [45]). We have calculated summary measures as describe in the following. The Magnitude of Relative Error can be defined as:

$$MRE = |SIZE_{real} - SIZE_{pred}| / SIZE_{real} \quad (2)$$

where $SIZE_{real}$ and $SIZE_{pred}$ are the actual and the predicted sizes, respectively. MRE has been calculated for each point in the data set. All the MRE values were aggregated across all the data points using the mean and the median, giving rise to the Mean of MRE (MMRE) and the Median MRE (MdMRE).

The prediction at level l [10], defined as:

$$Pred(l) = k / N \quad (3)$$

where k is the number of data points whose MRE is less than or equal to l , and N is the total number of data points. $Pred(l)$ is a quantification of the percentage of predictions whose error is less than $l\%$. In the context of effort estimation, where these measures were proposed [10], l is widely set to 0.25 and a good estimation model should have a $MMRE \leq 0.25$ and $Pred(0.25) \geq 0.75$, that is, the mean estimation error should be less than 25%, and at least 75% of the estimated values should fall within 25% of their actual values [10]. In the current study, which can be considered a preliminary investigation, we decided to use $l=0.25$. In the future, we will further analyze this point.

Since MMRE and Pred(0.25) could fail in fully representing a phenomenon as argued in [31], [32], we complemented these indicators with the analysis of the boxplots of absolute residuals. Boxplots are widely employed in exploratory data analysis since they provide a quick visual representation to summarize the data, using five values: median, upper and lower quartiles, minimum and maximum values, and outliers [32]. We used boxplots to graphically render the spread of the absolute residuals, calculated as $|SIZE_{real} - SIZE_{pred}|$. Moreover, we tested the statistical significance of the obtained results, by using Wilcoxon test on absolute residuals, in order to establish if one of employed estimation models provided better results than the others [32], [46]. In particular, we verified the following null hypothesis: “the two considered populations have identical distributions”. This kind of test is used to verify the hypothesis that the mean of the differences in the pairs is zero. In all the performed statistical tests we decided, as usual, to accept a probability of 5% of committing a Type-I-Error [49], i.e., of rejecting the null hypothesis when it is actually true.

IV. RESULTS AND DISCUSSION

A. Relationship between FP and T

In order to apply LR we first verified the normality of distributions (i.e., FP and T). In particular, we applied the Shapiro test [44], by considering as null hypothesis the normality of the distributions. The test results revealed that FP and T distributions were not normally distributed since the p-values were less than 0.05 for almost all the considered data sets (see Table II). So, as usual for this kind of study (see e.g., [5], [6], [30]), we decided to apply a Log transformation of the variables.

Moreover, the application of LR requires the verification of the following assumptions: linearity (i.e., the existence of a linear relationship between the independent variable and the dependent variable); homoscedasticity (i.e., the constant variance of the error terms for all the values of the independent variable); and residual normality (i.e., the normal distribution of the error terms).

TABLE II. NORMALITY OF DISTRIBUTIONS

SHAPIRO		
TEST	FP	T
t-statistic	0.745	0.832
p-value	<0.01	<0.01

Linearity. The Pearson's correlation test revealed that there was a linear relationship between the dependent (FP) and the independent variables (T) involved in the model (p-value = <0.01) [18].

Homoscedasticity. We investigated the homoscedasticity assumption by performing a Breush-Pagan Test [1], with the homoscedasticity of the error terms as null hypothesis. The assumption can be considered to be verified since the p-

values (0.349) was greater than 0.05 and thus the null hypothesis cannot be rejected.

Residual normality. To verify the normality assumption, we used the Shapiro-Wilk Test, by considering as null hypothesis the normality of error terms. The results of the test revealed that the assumption can be considered to be verified since the p-value (0.321) was greater than 0.05 and thus the null hypothesis cannot be rejected.

It is worth noting that we also verified the presence of extreme values that might unduly influence the models obtained from the regression analysis. As designed in Section III.B, we used Cook's distance to identify possible influential data points. In particular, the data points with a Cook's distance higher than 0.24 (= 4/17, where 17 is the number of data points) were removed from the corresponding training set in order to test the model stability. The analysis of outliers revealed that data point 17 (Cook's distance = 0.31), 2 (Cook's distance = 0.46), and 7 (Cook's distance = 0.68) may be influential data points. New models were generated without these data points. In the new models the independent variable (i.e., T) remained significant, the R^2 improved a little, and the coefficients present similar value to the ones in the previous models. Thus, these data points were not removed [38].

Table III presents the results of the LR analysis with statistics on useful indicators to verify the quality of the obtained models.

TABLE III. THE RESULTS OF THE LR ANALYSIS USING T AS INDEPENDENT VARIABLE

	Value	Std.Err	t-value	p-value
Intercept	2.715	0.238	11.39	<0.01
Coefficient of T	0.850	0.078	10.88	<0.01
	R^2	Std.Err	F	Sign F
	0.888	0.222	118.5	<0.01

We can observe that the linear regression analysis was successfully applied. Indeed, the obtained model is characterized by a high R^2 value (89%), that is the amount of the variance of the dependent variable FP that is explained by the model related to the variable T. Furthermore, a high F value (118.5) and a low p-value (<0.01) are obtained, indicating that the prediction is possible with a high degree of confidence. The t-values and p-values for the coefficient and the intercept are greater than 1.5 and less than 0.05, respectively, meaning that T is a significant predictor with a confidence of 95%.

The Equation as read from the final model's output is:

$$\ln FP = 0.850 \cdot \ln T + 2.715 \quad (1a)$$

which, when transformed back to the raw data scale, is:

$$FP = 15.105 \cdot T^{0.850} \quad (1b)$$

As designed in Section III.B, to evaluate the size estimation accuracy we performed a leave-one-out cross validation, whose results are reported in Table IV. We can

conclude that T is a good indicator of functional size, when used in combination with LR, since MMRE and MdMRE values are less than 0.25. Pred(0.25) value is quite close to 0.75. These results suggest that the software company involved in our study could obtain application size estimations just exploiting information on physical tables, with an error of about 21%.

TABLE IV. ACCURACY RESULTS FOR T BASED MODEL

MMRE	MdMRE	Pred(0.25)
0.21	0.18	0.71

B. Relationship between FP and R and R/W

In this section we report on the analysis performed to investigate the contribution of R and R/W to functional size. Similar to the analysis performed in the previous section, before applying linear regression we carried out a Log transformation since the Shapiro test revealed that FP, R, and R/W distributions were not normally distributed.

We first verified the linear regression assumptions, i.e., linearity, homoscedasticity, and residual normality. The Pearson's correlation test revealed a linear relationship between FP and R (p-value = <0.01), while for R/W the correlation was not so strong (p-value = 0.07). As for homoscedasticity, we applied the Breush-Pagan Test [1]. The results revealed that the assumption can be considered to be verified (p-value = 0.973). The residual normality assumption was confirmed by the Shapiro-Wilk Test (p-value = 0.994). The analysis of outliers revealed the presence of influential data points (extreme values that unduly affected the estimation model), which were removed from the dataset.

It is worth to point out that the application of MSWR revealed that the best fitting model was the one including both R and R/W as independent variables.

Table V presents the results of LR and MSWR with statistics on useful indicators to verify the quality of the obtained model. We can observe that the model is characterized by a high R^2 since about 94% is the amount of the variance of the dependent variable FP that is explained by the model related to the variables R and R/W. Moreover, the coefficients of R and R/W were characterized by a p-value less than 0.05 and a t-value greater than 1.5, meaning that they are significant predictors with a confidence of 95%.

The Equation as read from the final model's output is:

$$\ln FP = 3.048 + 0.486 \cdot \ln R + 0.534 \cdot \ln R/W \quad (2a)$$

which, when transformed back to the raw data scale, is:

$$FP = 21.073 \cdot R^{0.486} \cdot R/W^{0.534} \quad (2b)$$

The leave-one-out cross validation performed to evaluate the size estimation accuracy produced MMRE = 0.11, MdMRE = 0.12, and Pred(0.25) = 0.73 (Table VI). We can conclude that also R and R/W can be profitably exploited to

estimate the functional size, when used in combination with linear regression, since MMRE and MdMRE values are less than 0.25. Furthermore, Pred(25) value is quite close to 0.75. In particular, the software company involved in our study could obtain application size estimations that should be characterized by an error less than 12% just exploiting information on the number of read and read/written tables.

TABLE V. THE RESULTS OF THE LR ANALYSIS USING R AND R/W AS INDEPENDENT VARIABLES

	Value	Std.Err	t-value	p-value
Intercept	3.048	0.209	14.563	<0.01
Coefficient of Log R	0.486	0.048	10.563	<0.01
Coefficient of Log R/W	0.534	0.084	6.371	<0.01
R^2	0.939			
Std.Err	0.172			
F	77.71			
Sign F	<0.01			

TABLE VI. ACCURACY RESULTS FOR R AND R/W BASED MODEL

MMRE	MdMRE	Pred(0.25)
0.11	0.12	0.73

C. Comparison of the obtained models

The analysis has revealed that T, R, and R/W are significant predictors of functional size and the estimation models are characterized by an MMRE less than 21% in the worst case. Thus, we can positively answer our research questions, i.e., RQ1 and RQ2.

We can also observe that the higher R^2 and the lower MMRE were obtained with R and R/W. However, the results of T are close. The analysis of boxplots in Figure 1 also suggests that the distributions of the two models are quite similar since the box length and tails of the two boxplots are quite similar and both have an outlier. However, we can note that the median of the boxplot for the R and R/W based model is more close to zero.

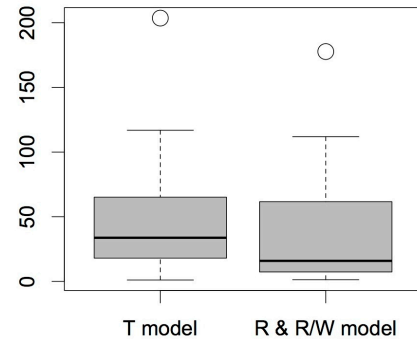


Figure 1. Fig. 1. Boxplot of absolute residuals

This is confirmed by the comparison between the obtained models performed through the analysis of the absolute residuals, looking for statistically significant differences

[32], [46]. The results of the Wilcoxon test revealed that there was no significant difference between the absolute residuals obtained with the two models (p -value = 0.643).

V. THREATS TO VALIDITY

The collection of the information to determine size measures represents a crucial aspect and can bias the construct validity. The measurement task (of development effort and functional size) is included in the software life cycle for each project developed by the company involved in our study. Thus, we are sure that the collection of information is performed in a uniform fashion. Reliability of the data and lack of standardization should be taken into account for the internal validity [32][33][39]. The questionnaires used to collect the information on the employed measures were the same for all the applications and the project managers were instructed to correctly provide the required information. Instrumentation effects in general did not occur in this kind of studies. As for the conclusion validity, we carefully applied the estimation method and the statistical tests, verifying all the required assumptions. With regard to the external validity, it is recognized that the results obtained in an industrial context might not hold in other contexts [7].

VI. CONCLUSIONS

We have presented an empirical study performed to assess whether the number of physical tables and the number of read and read/written tables can be exploited to estimate the functional size of applications. We have employed 17 applications from InfoCamere, an Italian company, and exploited linear regression to build the estimation models.

The results revealed that both the T based model and the R and R/W based model provided good estimations of the functional size of the 17 applications involved in the study. Indeed, the models are characterized by an MMRE of 21% and 12%, respectively.

The results are considered interesting by the company providing the data since the asset of the software applications stock can be estimated just looking at specific components easy to obtain. Indeed, the main goal was to find a cheap approach to be employed as support for decisions of the board of directors and for attestations with stakeholders. Just to give an idea of the possible benefits, the Italian software company involved in our study needs about 4-6 hours to size a medium application by applying the traditional IFPUG Function Points Analysis (without considering the time spent to interview the people for searching the relevant information). On the other hand, the proposed approach based on the use of tables required about 30 minutes. It is important to observe that, before applying the proposed method based on physical tables, InfoCamere tried to apply the LOC based backfiring approach [26]. However it was considered not feasible by project managers due to several managerial issues, e.g., several programming languages were used to develop the operating software

applications and the tool employed for one language could not be employed for another one; a developer of the application to be measured had to be involved; and the source code of a complex application turned out to be hard to analyze.

Since replications of a study, in different settings and with larger data sets, are always required, as future work, we also plan to collect and analyze data from the same company as well as from others. Moreover, we would like to investigate if the proposed approach can be used also to estimate the size expressed in terms of other Functional Size Measurement methods, such as COSMIC.

Furthermore, we intend to compare our approach with more sophisticated ones, such as Indicative and Estimated NEMSA, and the four types of approximations proposed for COSMIC, i.e., Average Functional Process, Fixed Size Classification, Equal Size Bands, and Average Use Case [12]. Although they have been proposed with different goals, namely for estimating the size in the early phase of the development, they share with us the need to allow for an estimation of the functional size using a few information. Thus, we would like to analyze pros and cons of the different approaches not only with respect to the accuracy of the estimation but also with respect to the information required to apply the methods and then the time needed to carry out the measurement.

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